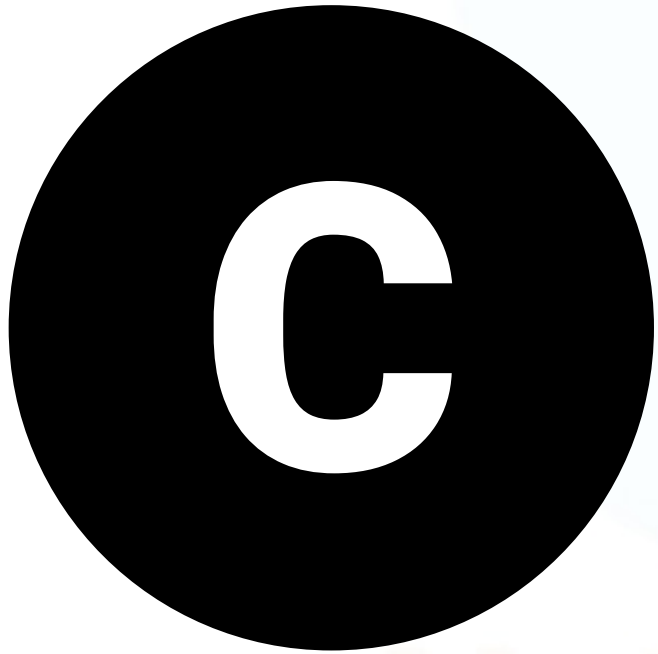
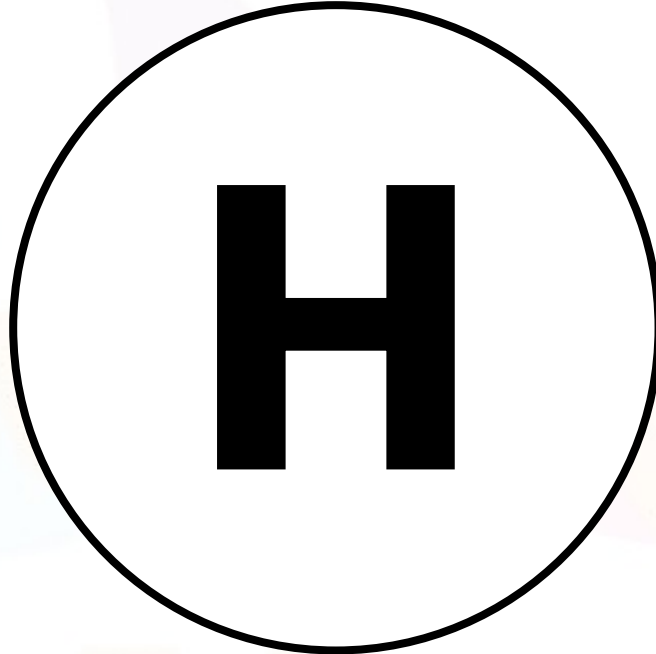


Organic = C-H / C-C bonds



Carbon

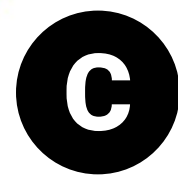


Hydrogen



Oxygen

Organic biomolecules



Nitrogen

Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



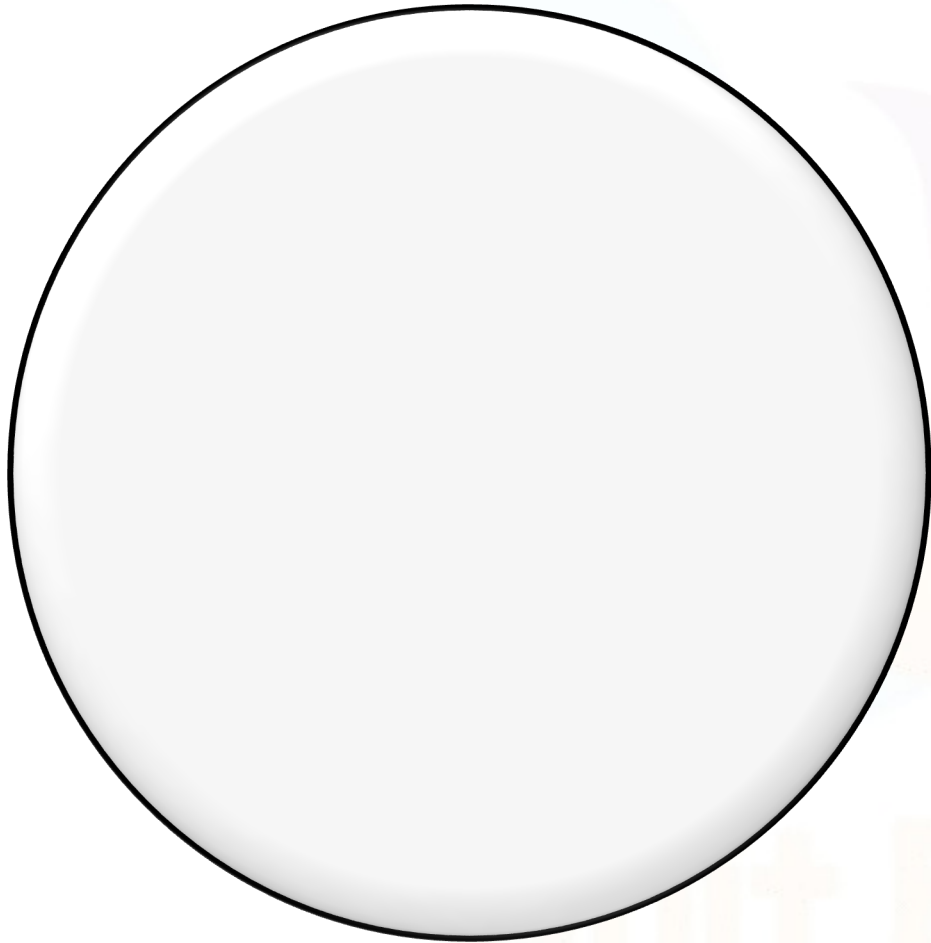
3. Proteins 蛋白質



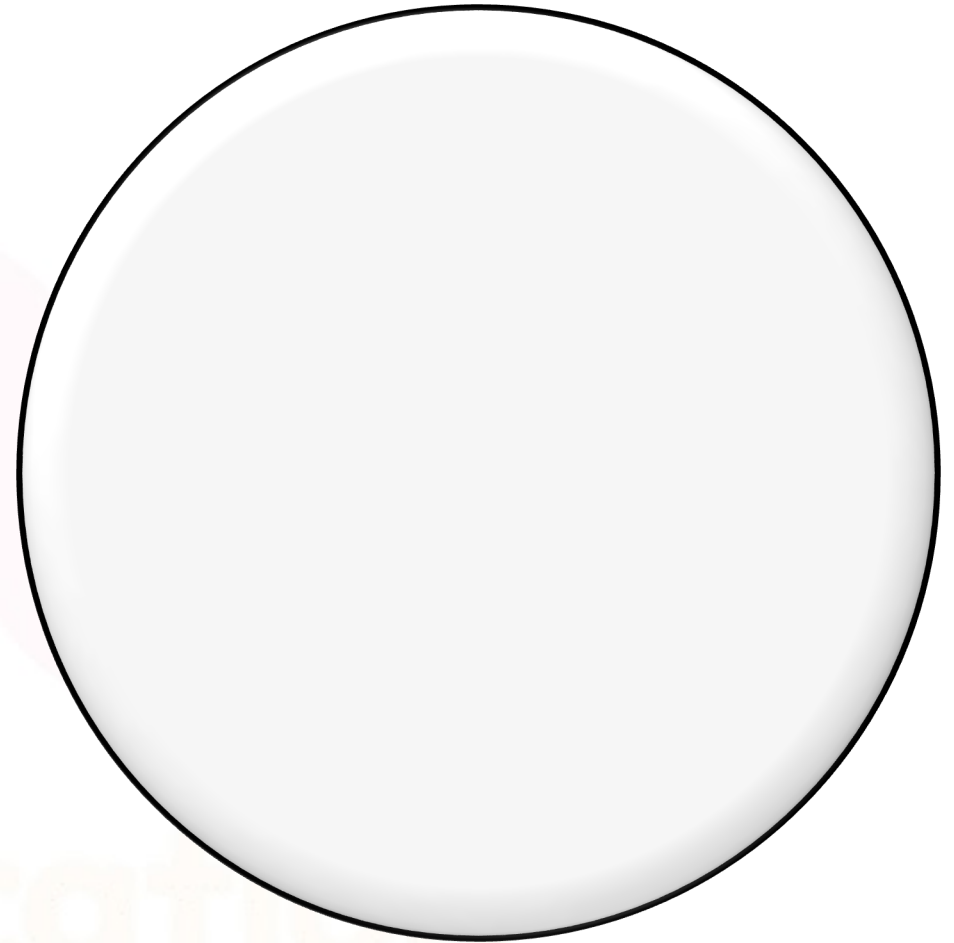
4. Nucleic acids 核酸



Condensation 縮合 and Hydrolysis 水解

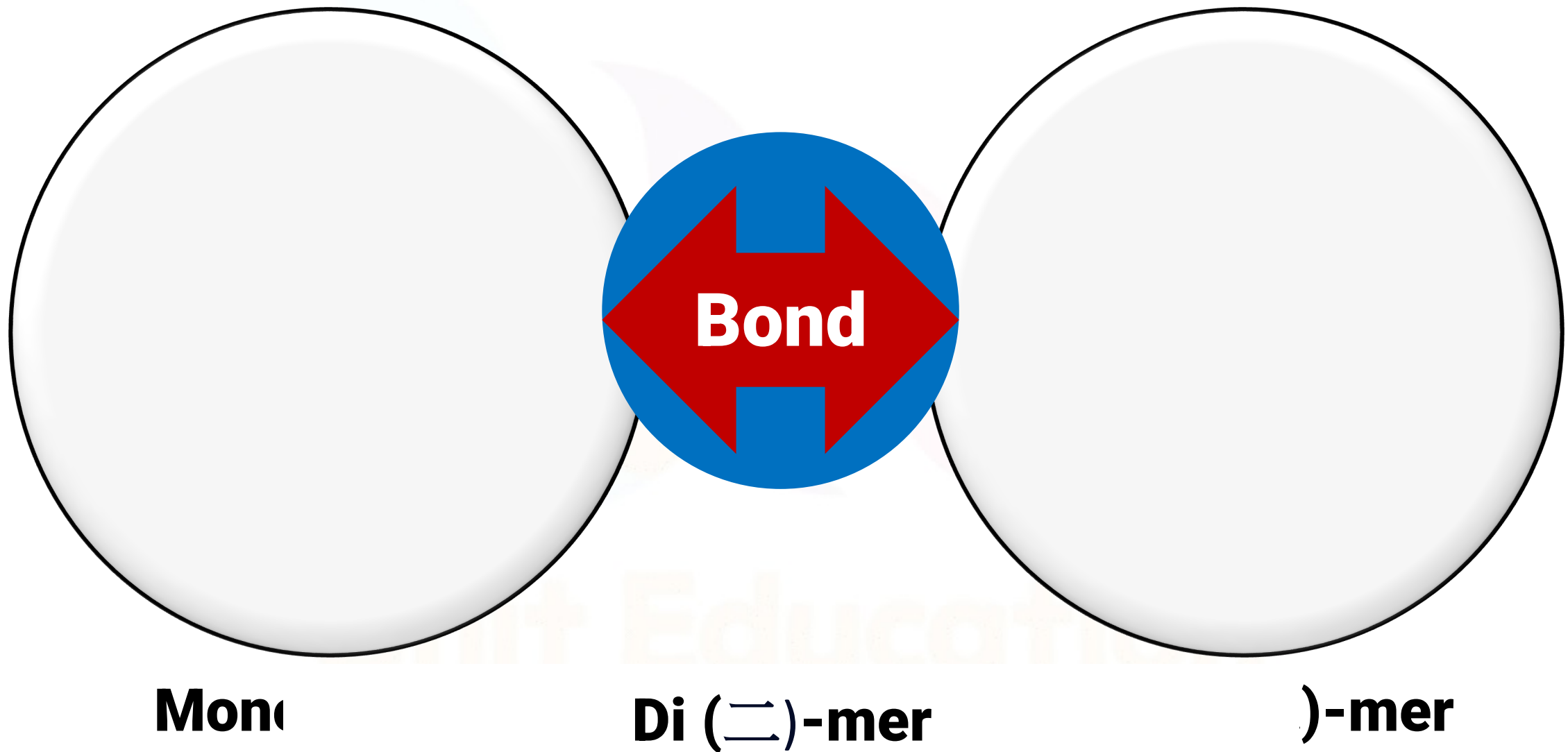


Mono單-mer

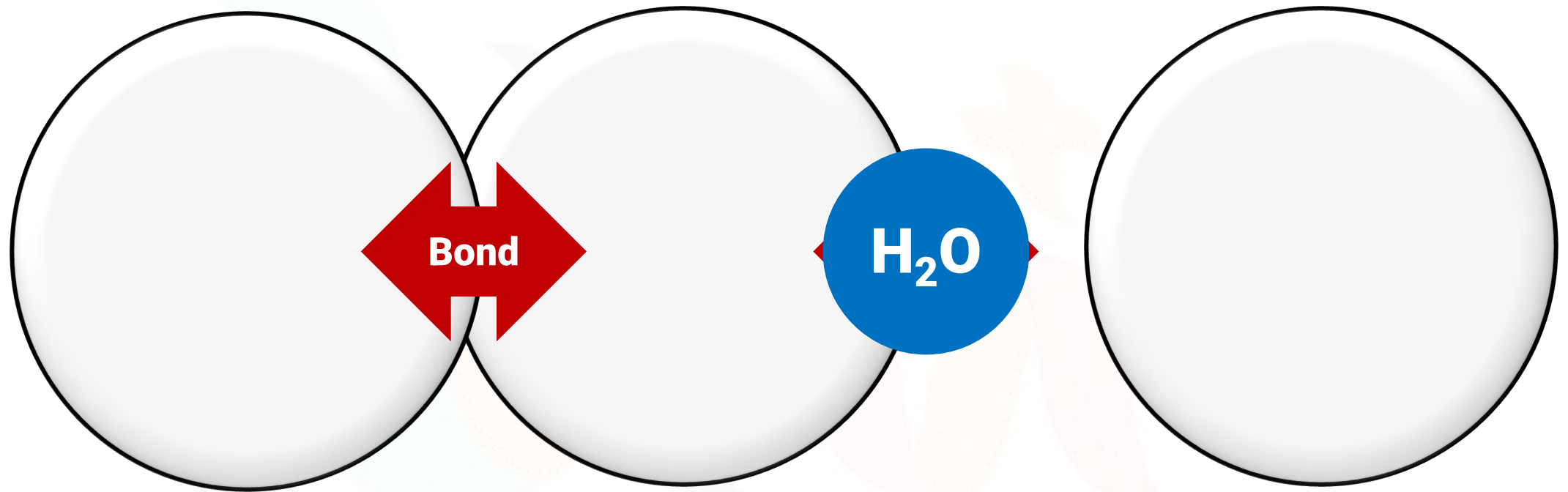


Mono單-mer

Condensation 縮合

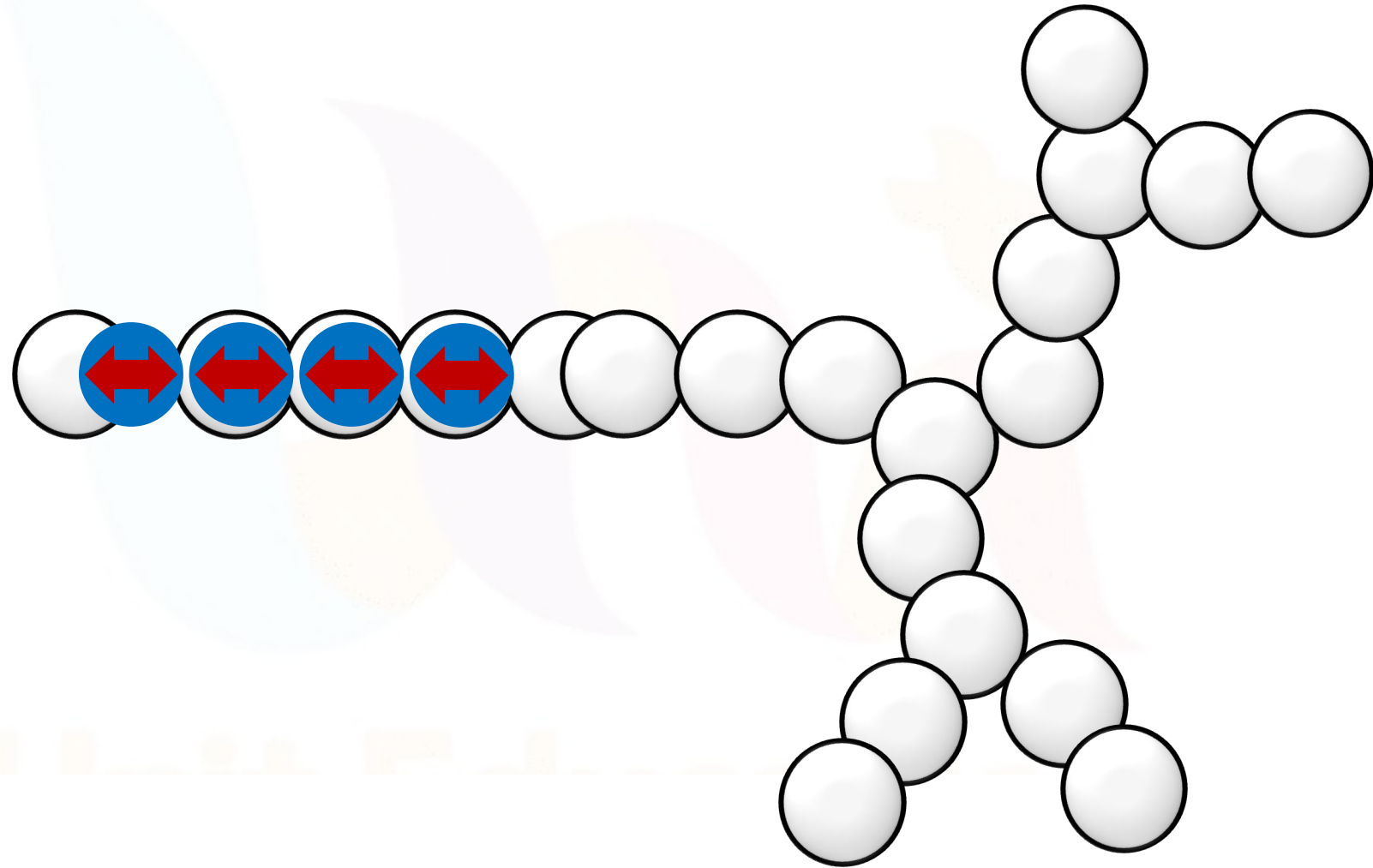


Condensation 縮合



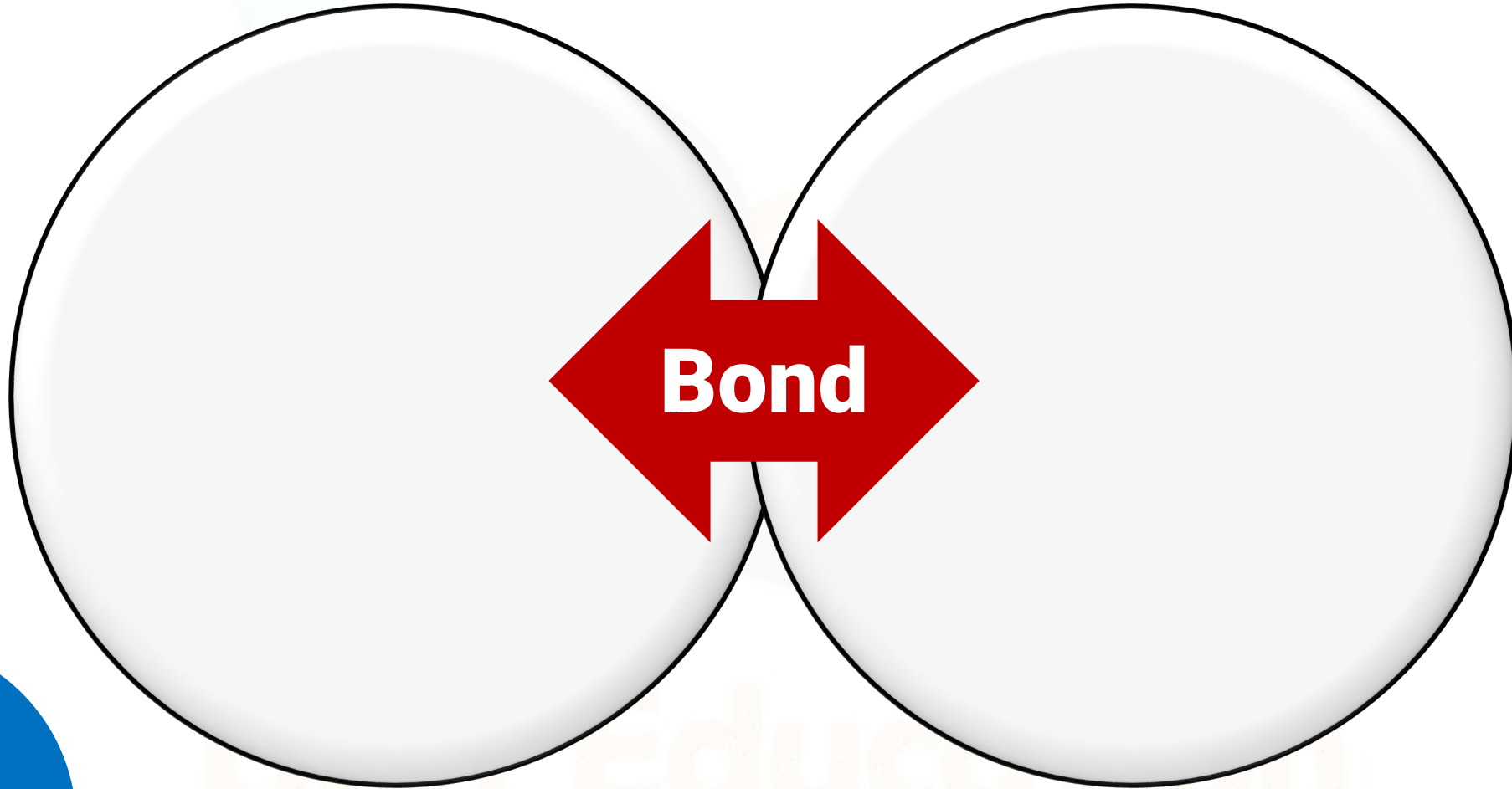
Poly (多)-mer

Condensation 縮合



Poly (多)-mer

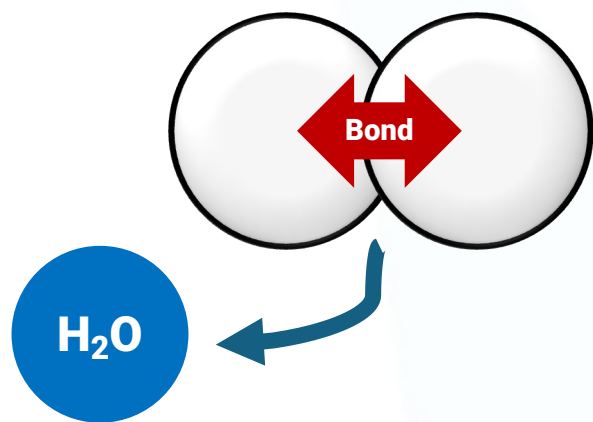
Hydrolysis 水解



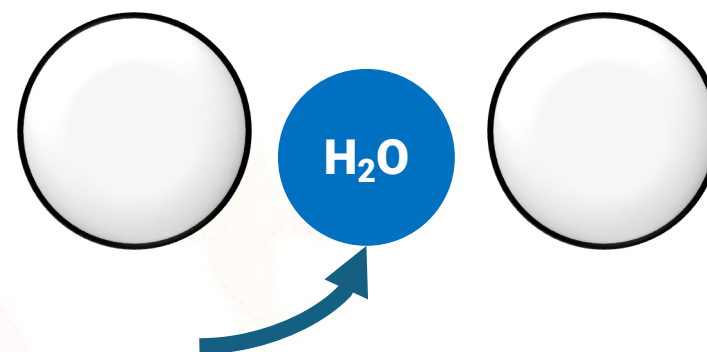
H_2O

Di (二)-mer

Condensation 縮合 and Hydrolysis 水解



“Condensation 係會出水”



“水解，用水去解 (bond)”

宜家加大炸雞餐有特價，但阿媽話唔健康唔比買；點樣用生物科概念說服佢炸雞餐有營養需要？



快樂水

總糖份: ~56g

碳水化合物(Carbohydrates)
係人不可或缺嘅**快速能量來源**
(Fast respiration fuel)
)，葡萄糖更係大腦必要嘅養份。



快樂條

總脂肪: ~20g

脂質(Lipids)係**高效嘅能量處**
存 (Efficient storage
)，亦係組成細胞膜同部份荷爾
蒙嘅重要成份。

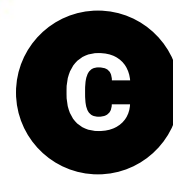


快樂雞

總蛋白質: ~32g

蛋白質(Proteins) 3D結構多變，
構成大部份身體組織
(Structural build blocks
)以及酶、抗體、血紅素等重要
分子。

Organic biomolecules



Nitrogen

Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質

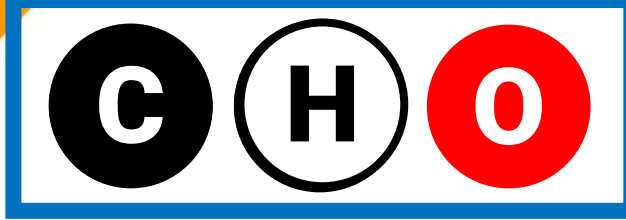


4. Nucleic acids 核酸



1. Carbohydrates

碳水化合物



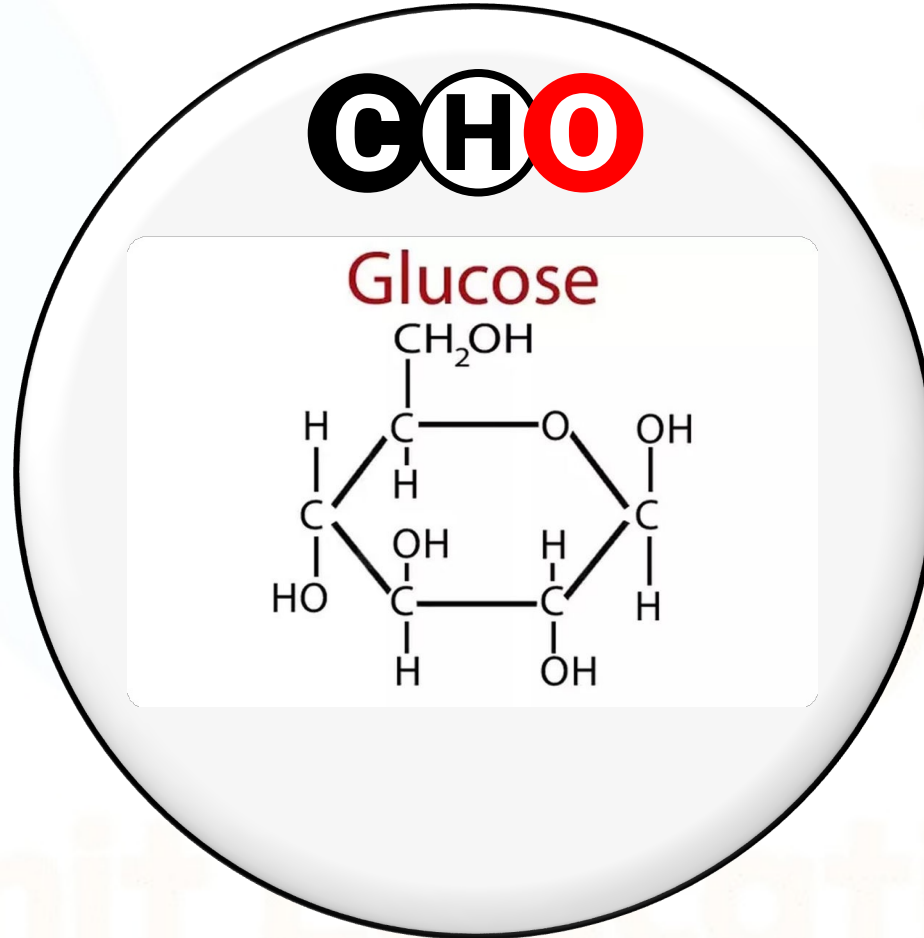
Nitrogen



Phosphorus



Sulfur



Monosaccharides 單糖

- Soluble in water
- Taste sweet
- Reducing sugars

Examples:

Glucose 葡萄糖

Fructose 果糖

Galactose 半乳糖

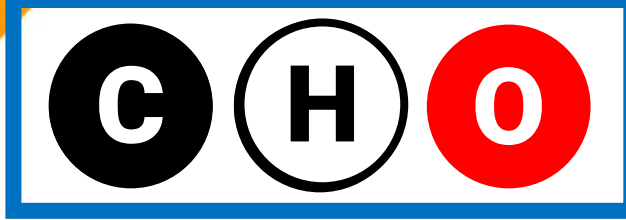
Ribose 核糖

Deoxyribose 脱氧核糖

Mono-saccharides (單糖)

1. Carbohydrates

碳水化合物



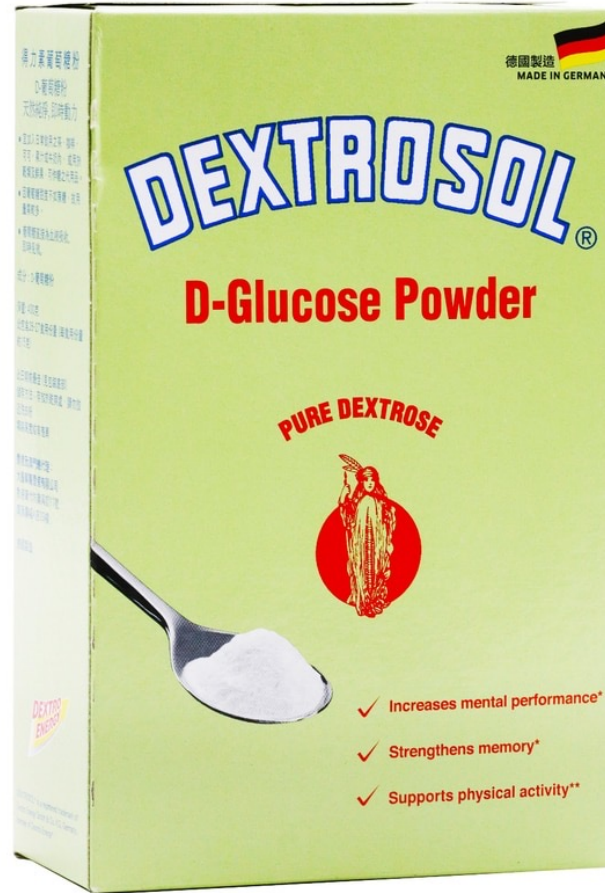
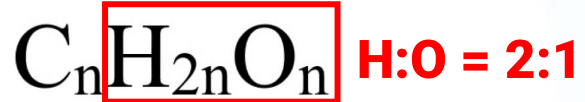
Nitrogen



Phosphorus



Sulfur



Monosaccharides 單糖

- Soluble in water
- Taste sweet
- Reducing sugars

Examples:

Glucose 葡萄糖

Fructose 果糖

Galactose 半乳糖

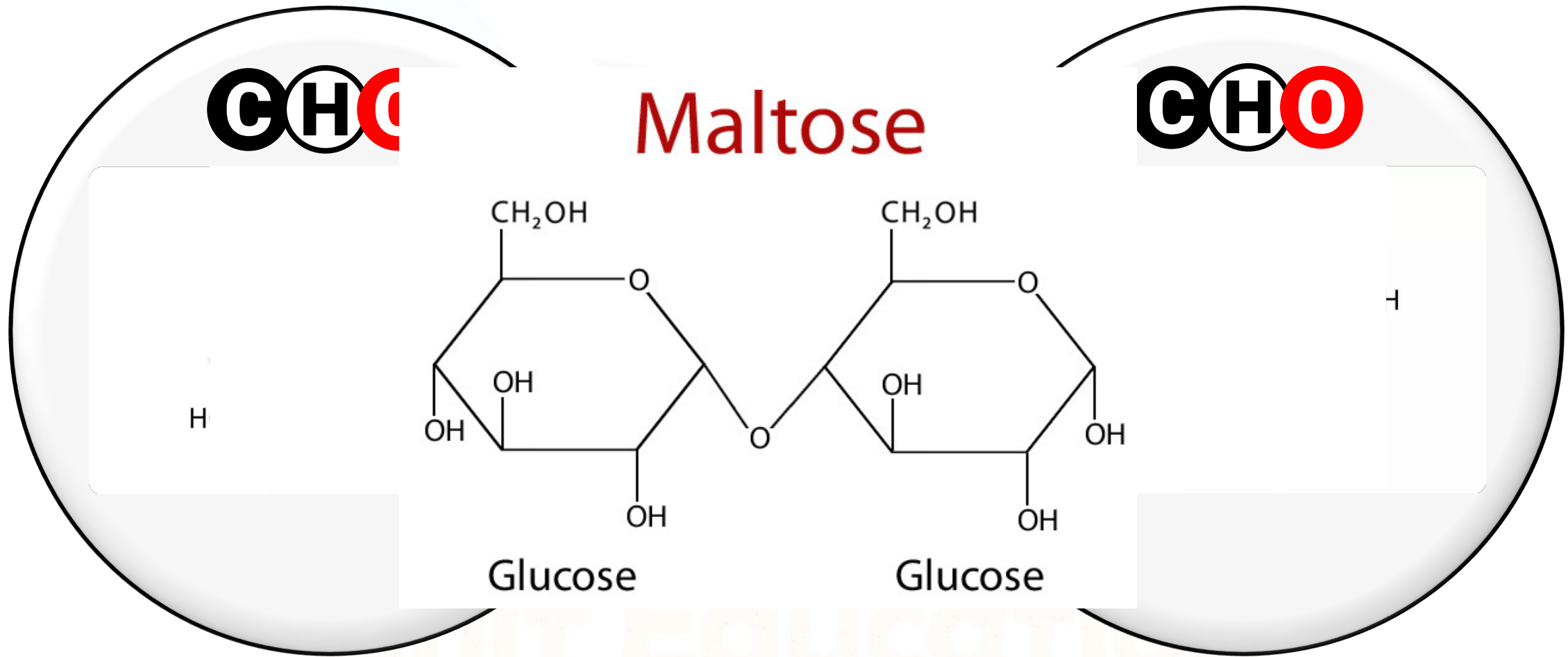
Ribose 核糖

Deoxyribose 脱氧核糖

Mono-saccharides (單糖)

1. Carbohydrates 碳水化合物

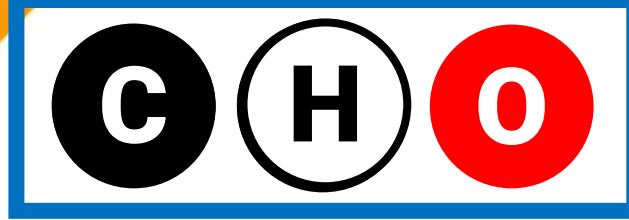
Condensation 縮合



Disaccharides (雙醣)

1. Carbohydrates

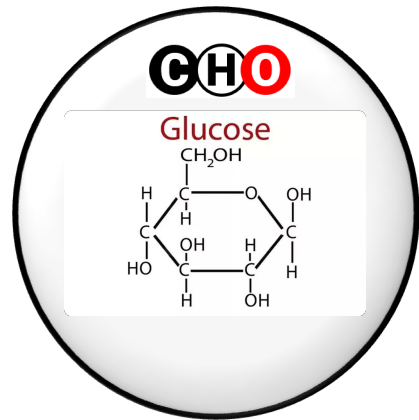
碳水化合物



Nitrogen

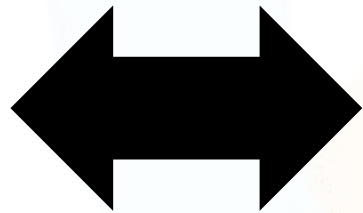
Phosphorus

Sulfur

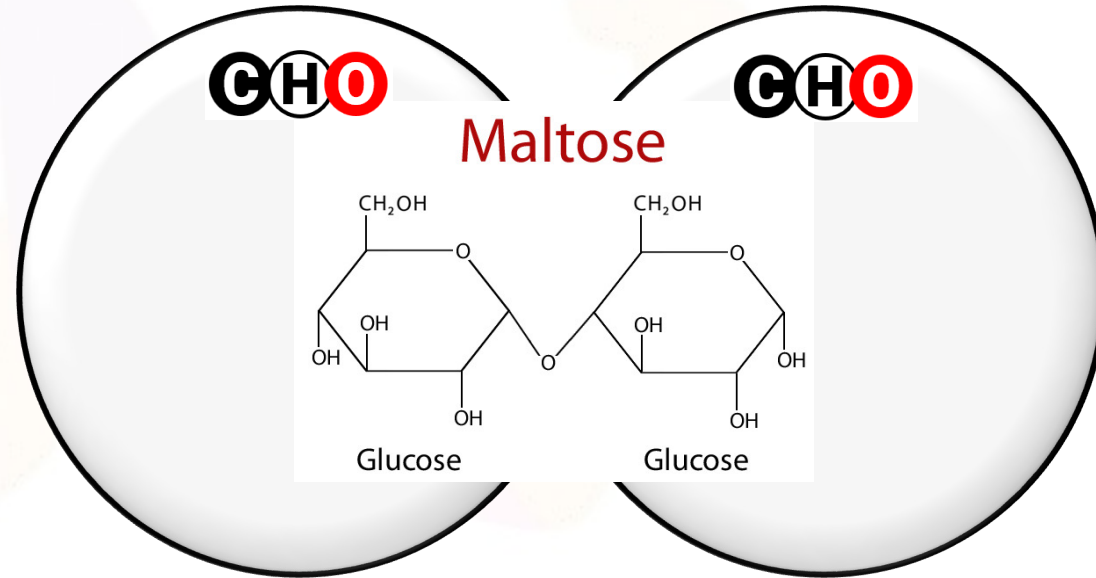


Monosaccharides
(單糖)

Hydrolysis



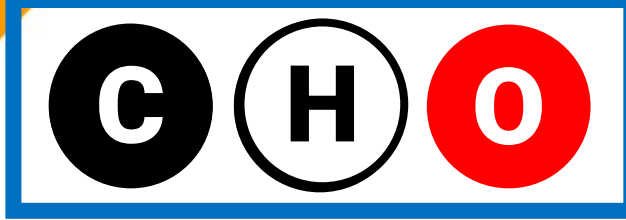
Condensation



Disaccharides (雙糖)

1. Carbohydrates

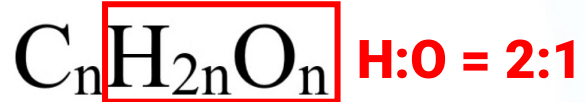
碳水化合物



Nitrogen

Phosphorus

Sulfur



Disaccharides (雙糖)

Disaccharides 雙糖

- Soluble in water
- Taste sweet
- Reducing sugars
(except sucrose)

Maltose 麥芽糖



Sucrose 蔗糖

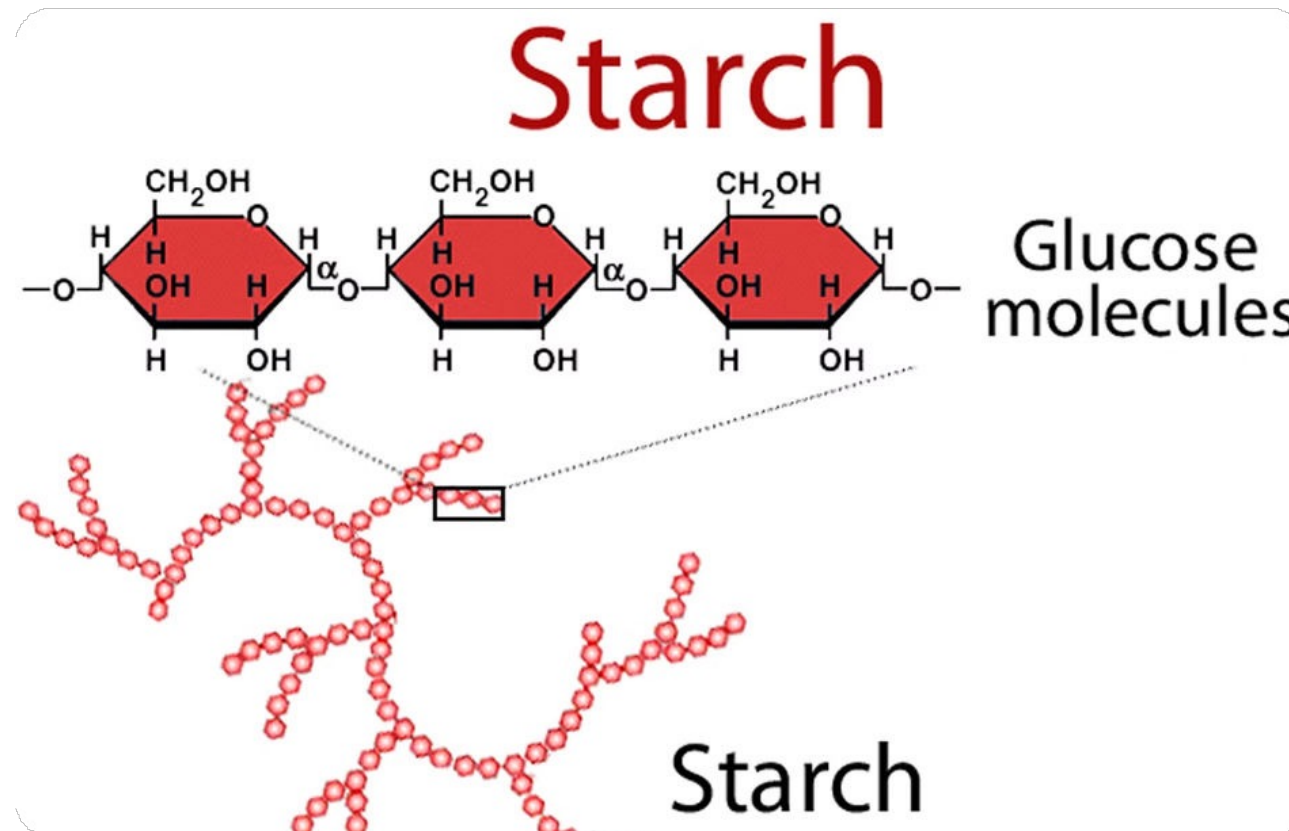


Lactose 乳糖



1. Carbohydrates 碳水化合物

Condensation 縮合



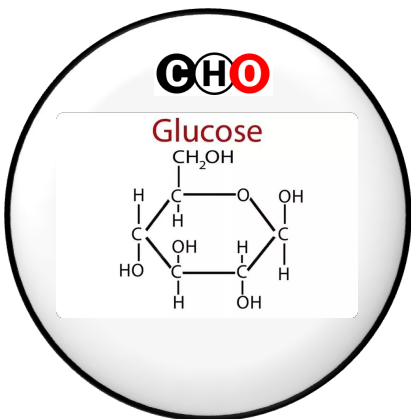
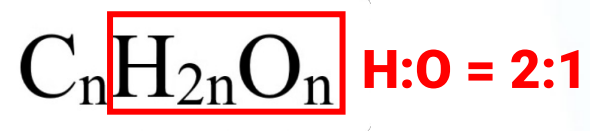
Polysaccharides (多醣)



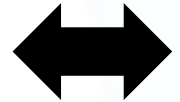
Nitrogen Phosphorus Sulfur

1. Carbohydrates

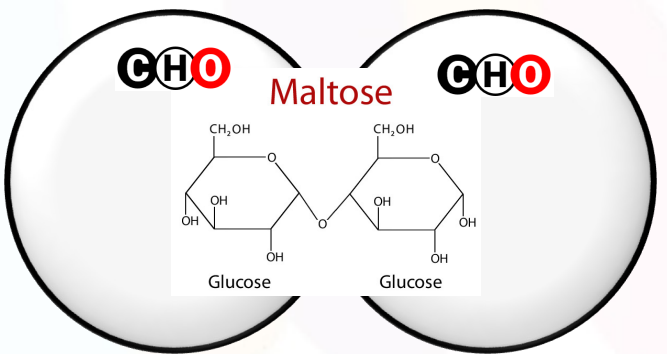
碳水化合物



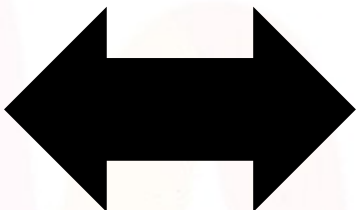
Hydrolysis



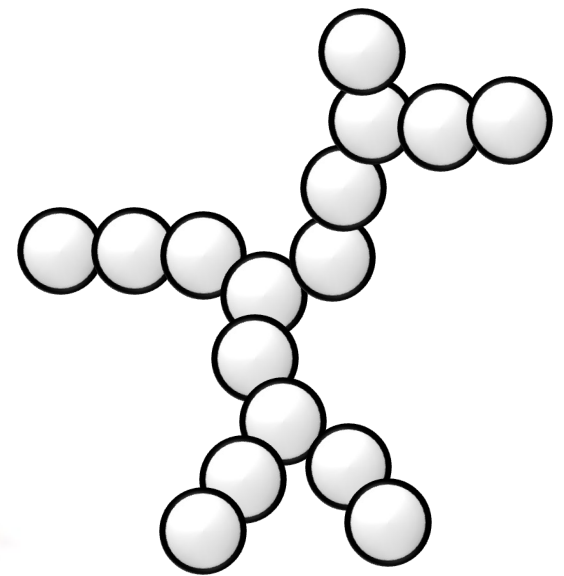
Condensation



Hydrolysis



Condensation



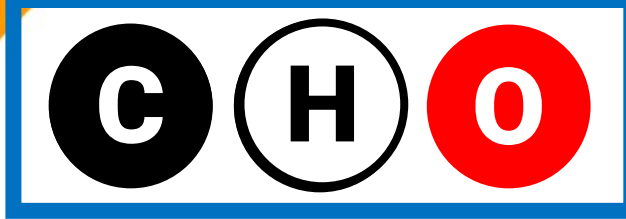
Monosaccharides
(單糖)

Disaccharides
(雙糖)

Polysaccharides (多糖)

1. Carbohydrates

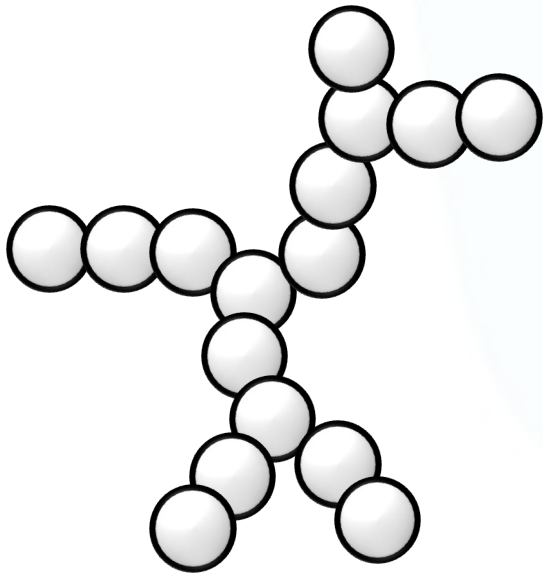
碳水化合物



Nitrogen

Phosphorus

Sulfur



Polysaccharides 多醣

- Insoluble in water
- Taste not sweet
- Non-reducing sugars
(except sucrose)

Examples:

Glycogen 肝糖

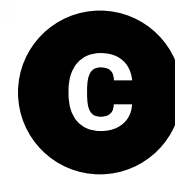
Cellulose 纖維素

Starch 澱粉

Polysaccharides (多醣)



Organic biomolecules



Nitrogen

Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質

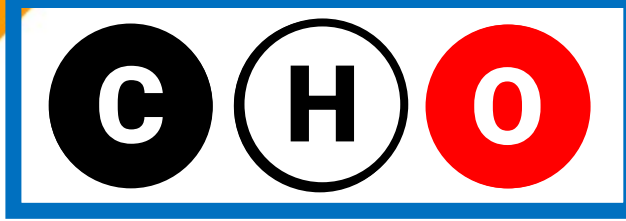


4. Nucleic acids 核酸



2. Lipids 脂質

$C_{51}H_{98}O_6$ **H:O > 2:1**



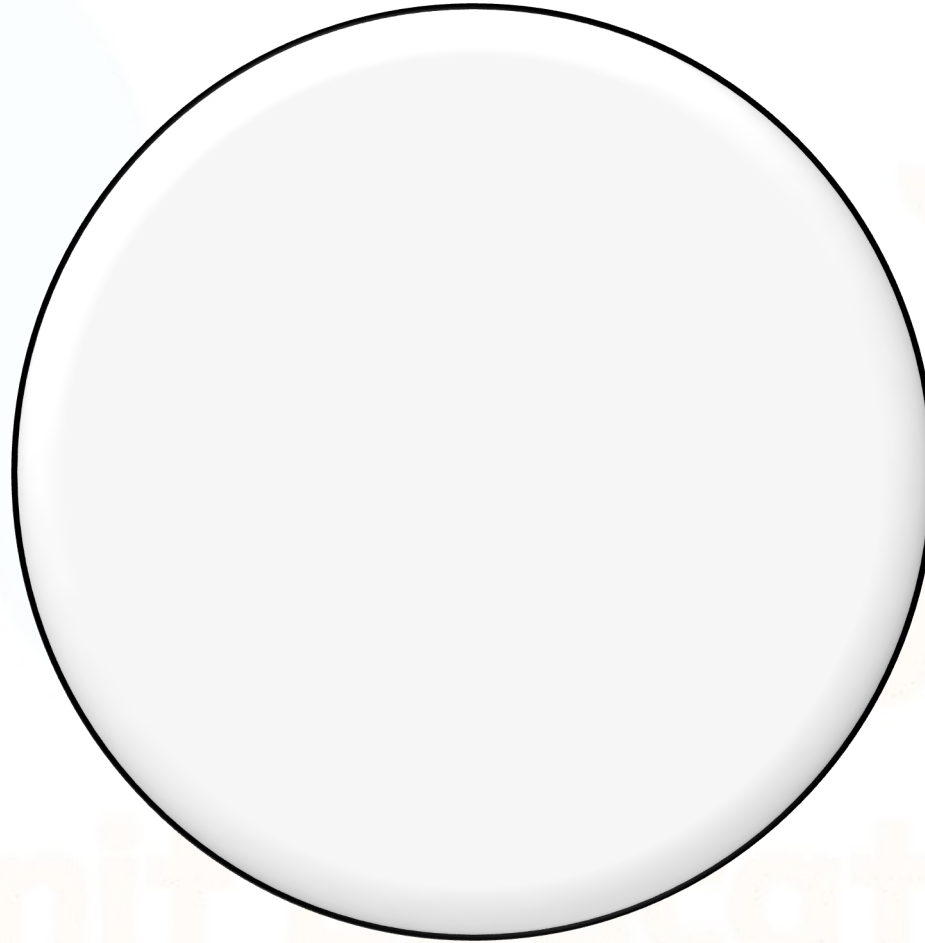
Nitrogen



Phosphorus

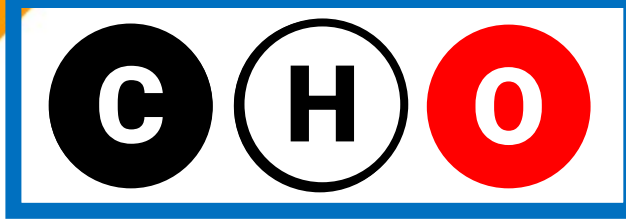
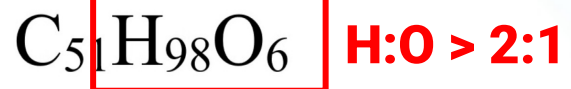


Sulfur



Mono單-mer

2. Lipids 脂質



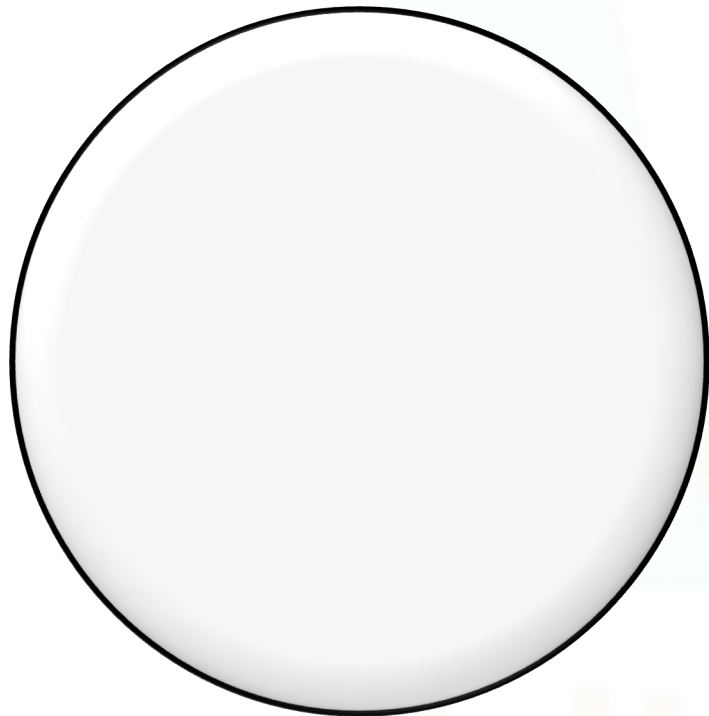
Nitrogen



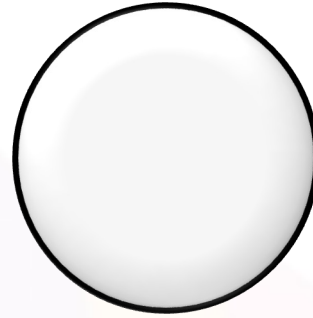
Phosphorus



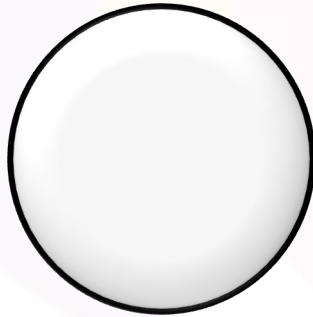
Sulfur



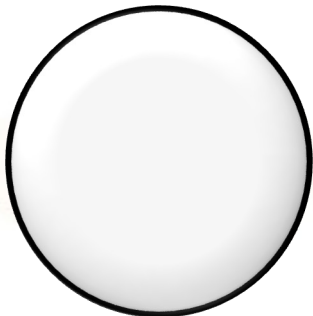
Glycerol (甘油)



Fatty acid (脂肪酸)



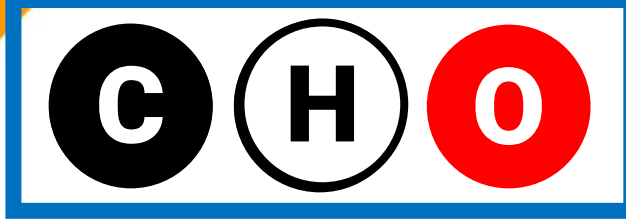
Fatty acid (脂肪酸)



Fatty acid (脂肪酸)

2. Lipids 脂質

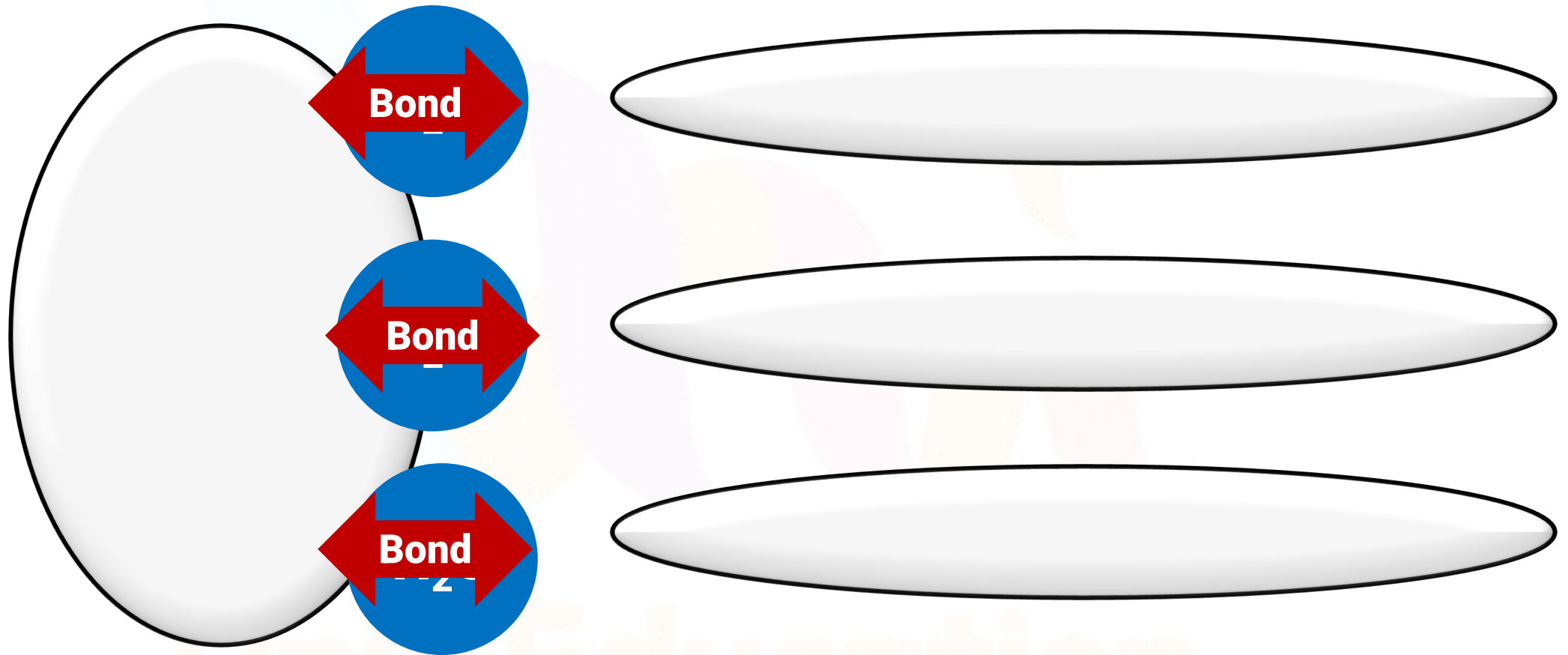
$C_{51}H_{98}O_6$ **H:O > 2:1**



Nitrogen

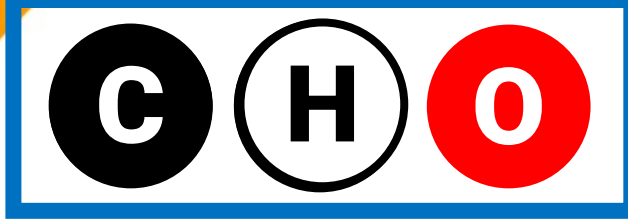
Phosphorus

Sulfur



3 Condensations 縮合!

2. Lipids 脂質



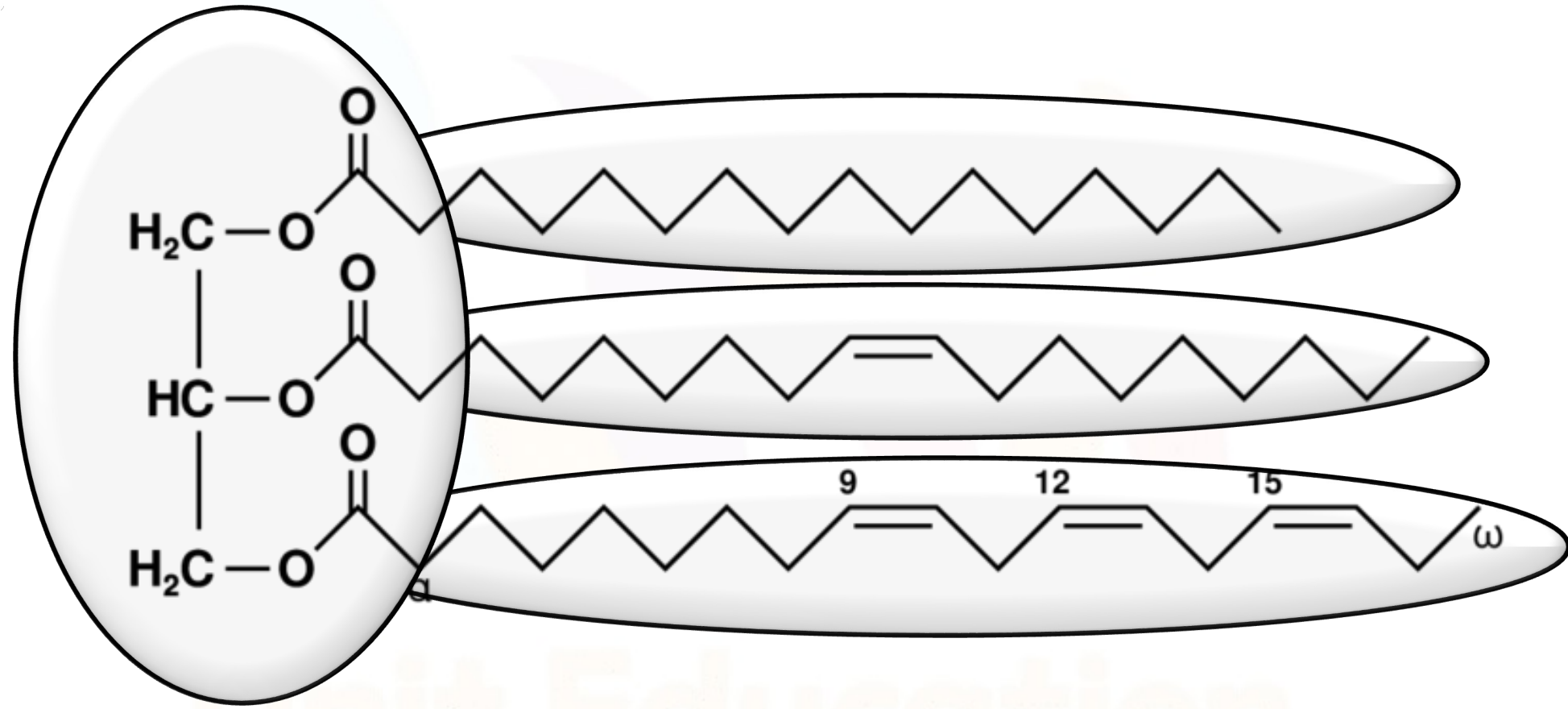
Nitrogen



Phosphorus

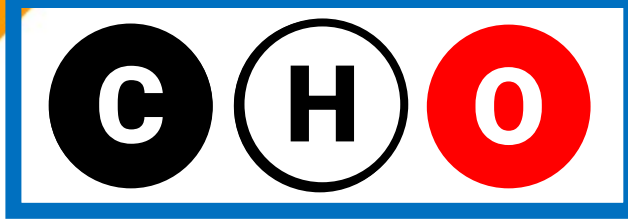


Sulfur



Tri-glycer-ides (甘油三酯)

2. Lipids 脂質



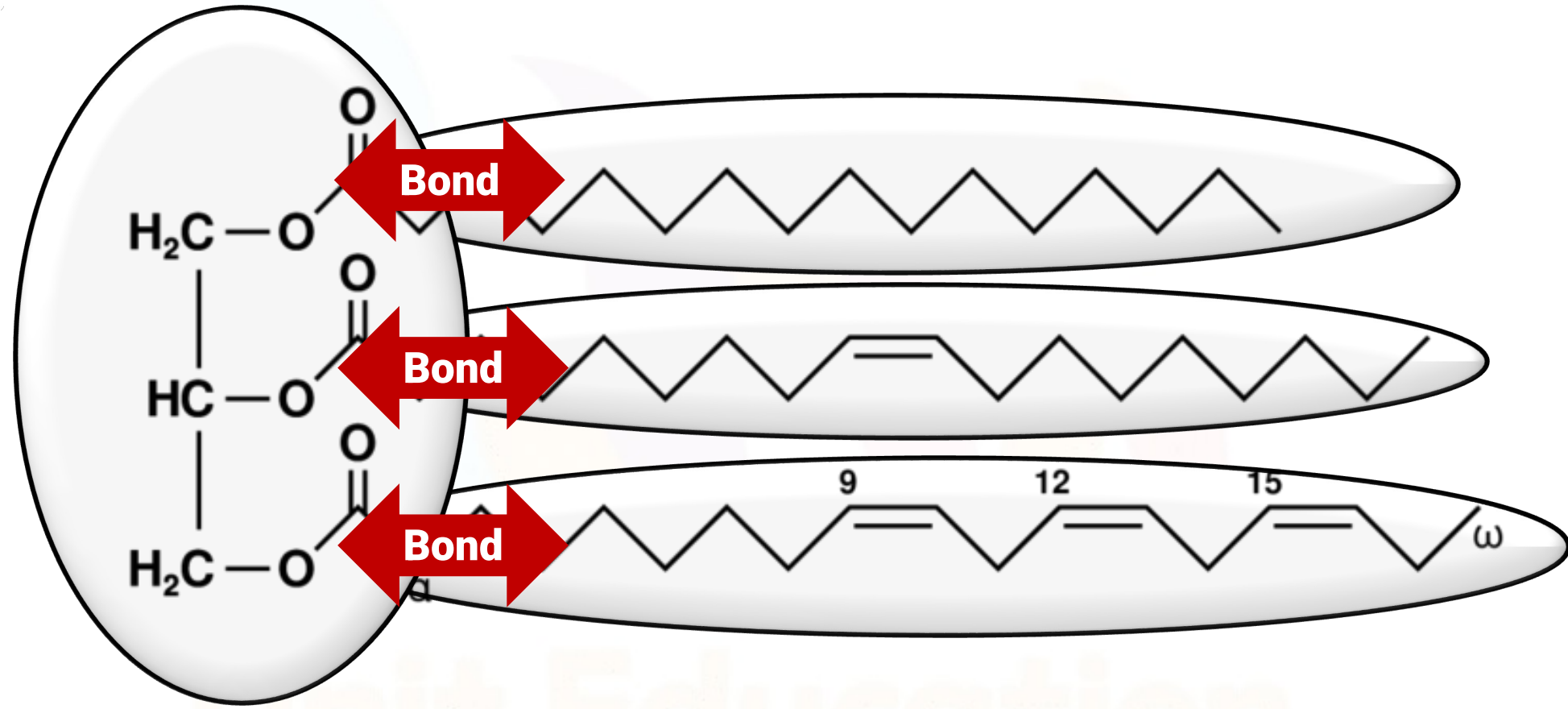
Nitrogen



Phosphorus

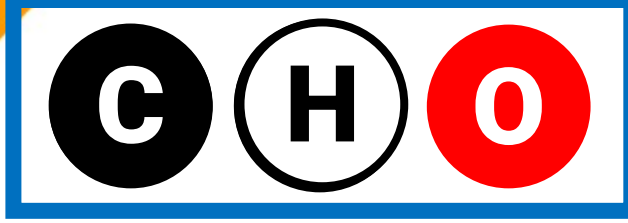


Sulfur



Triglycerides (甘油三酯)

2. Lipids 脂質



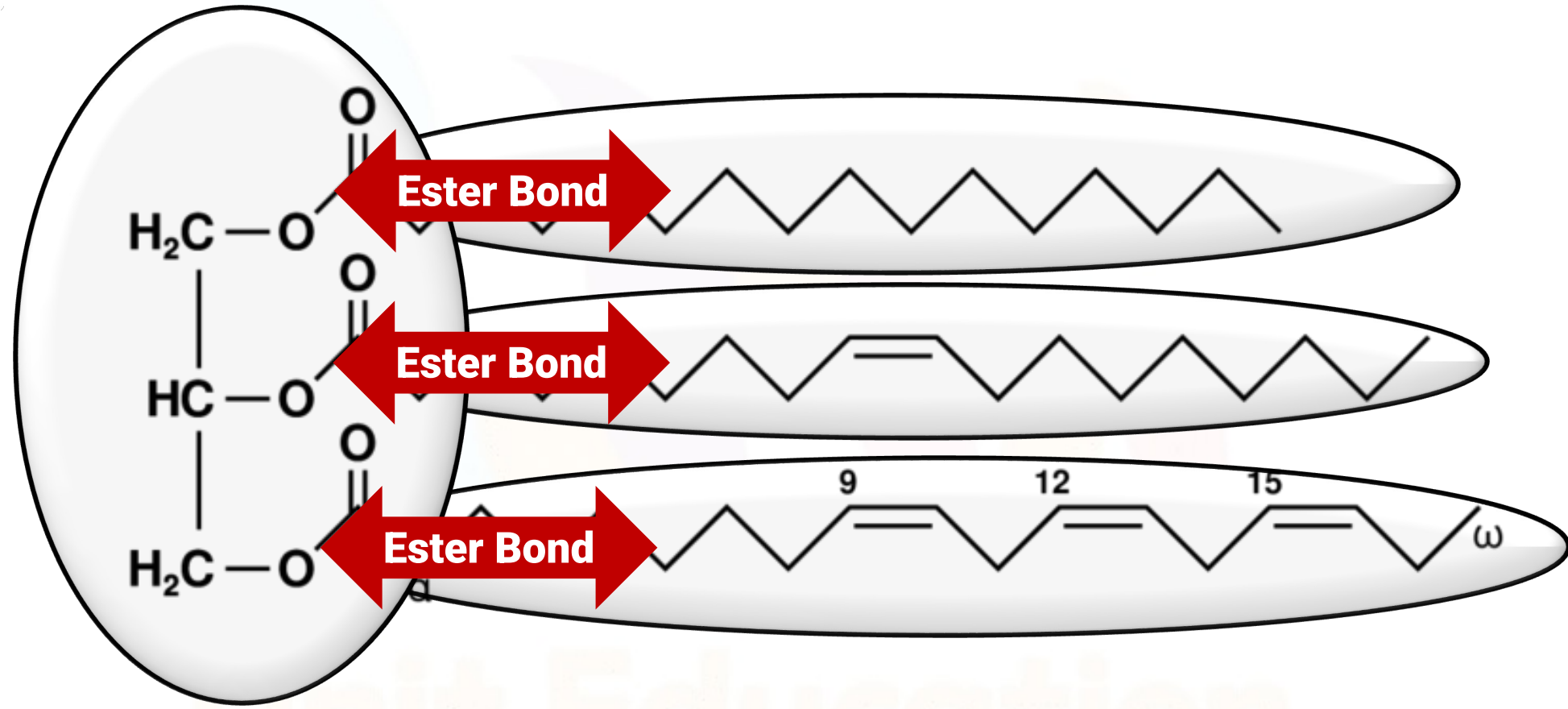
Nitrogen



Phosphorus

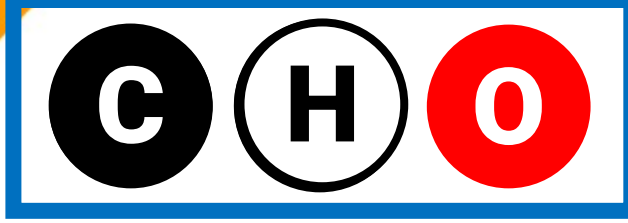


Sulfur



Triglycerides (甘油三酯)

2. Lipids 脂質



Nitrogen

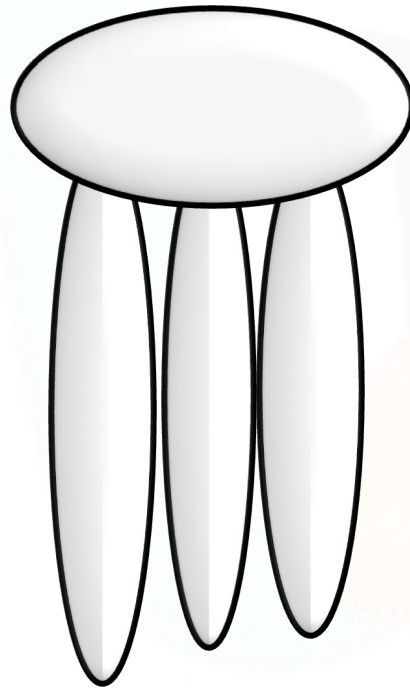


Phosphorus

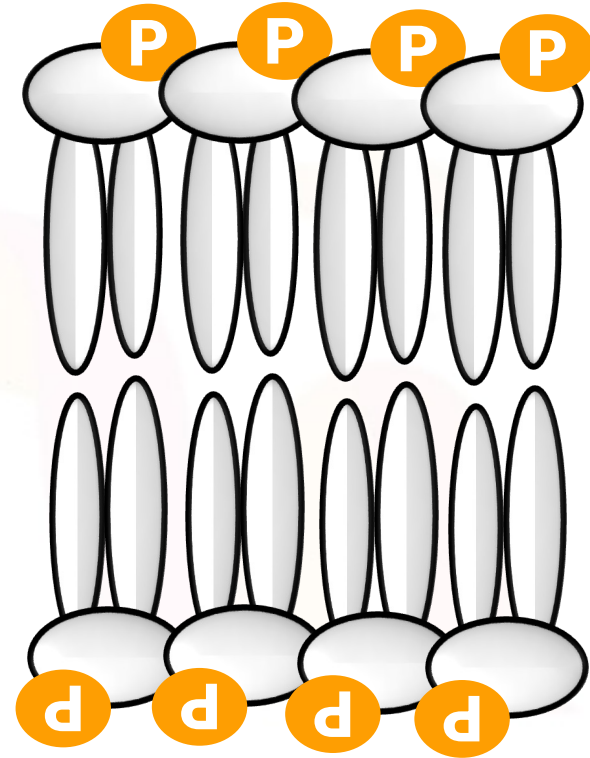


Sulfur

Triglycerides
(甘油三酯)



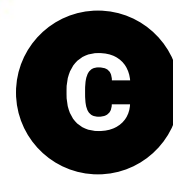
1 Glycerol (甘油)
+ 3 Fatty acids (脂肪酸)



Phospho-lipids
(磷脂)

1 Phosphate group
+ 1 Glycerol (甘油)
+ 2 Fatty acids (脂肪酸)

Organic biomolecules



Nitrogen

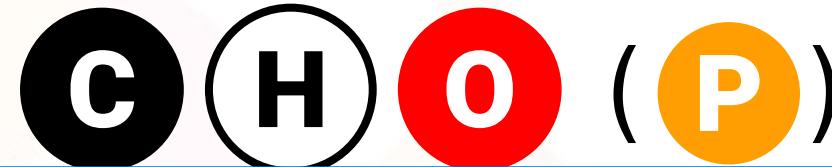
Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



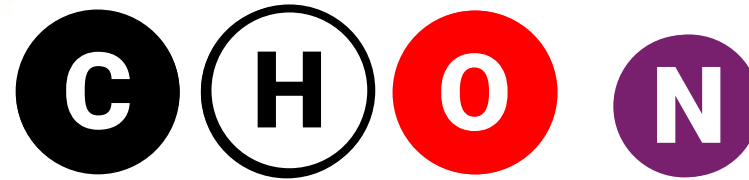
3. Proteins 蛋白質



4. Nucleic acids 核酸



3. Proteins 蛋白質



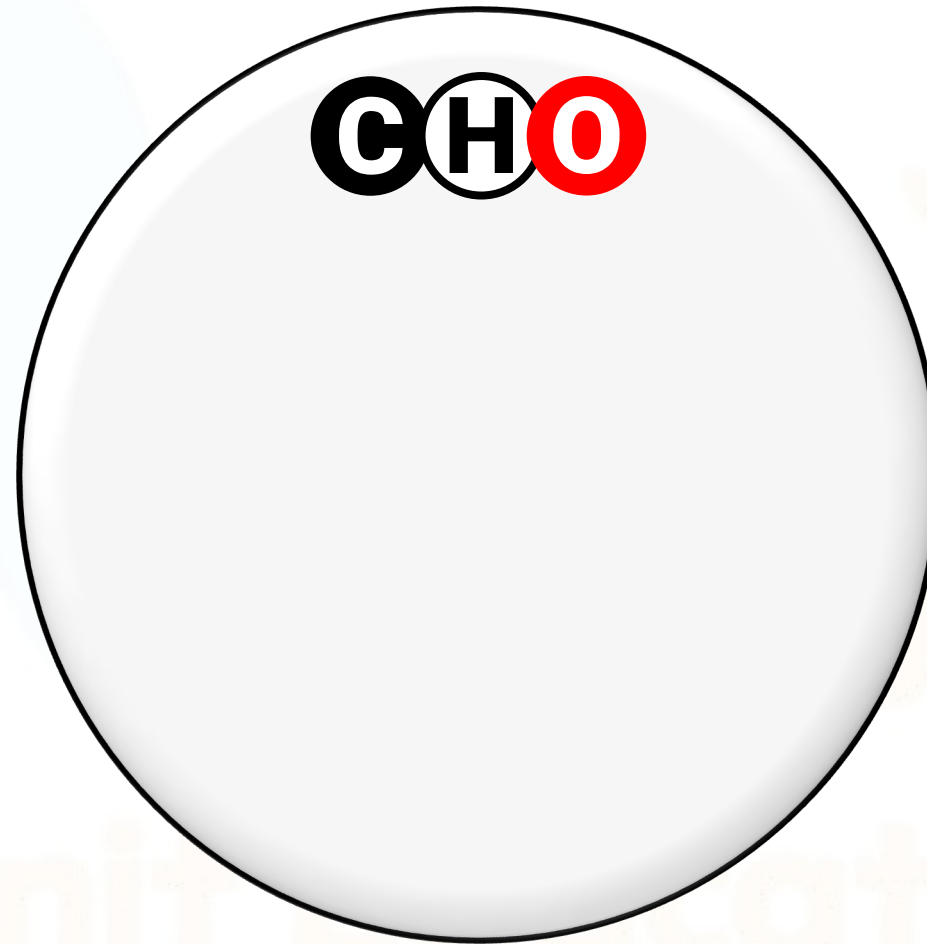
Nitrogen



Phosphorus

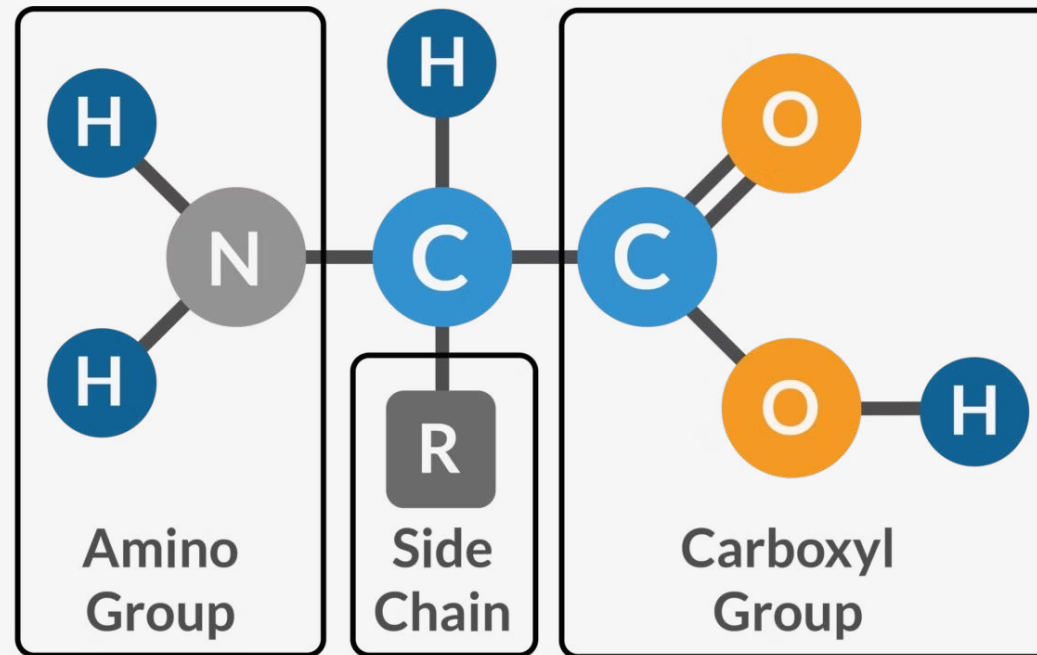
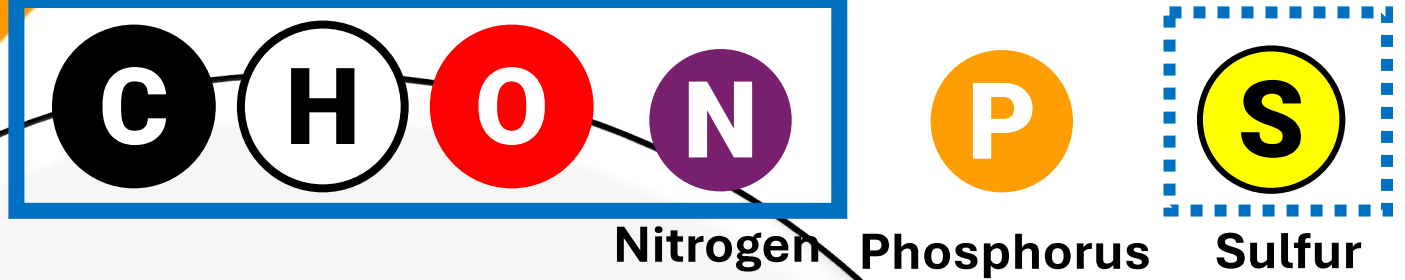


Sulfur



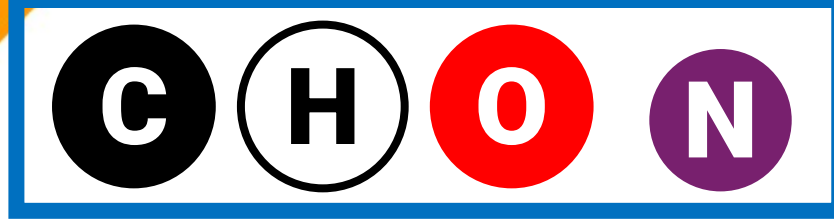
Amino acids (氨基酸)

3. Proteins 蛋白質



Amino acids (胺基酸)

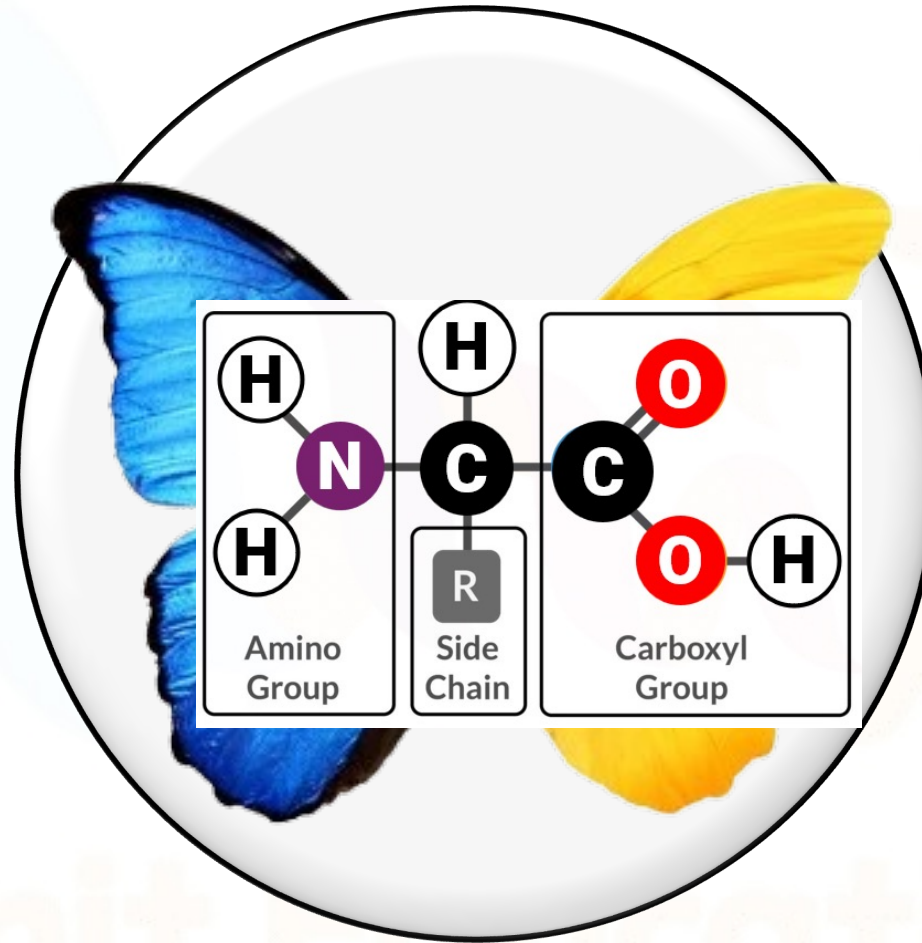
3. Proteins 蛋白質



Nitrogen

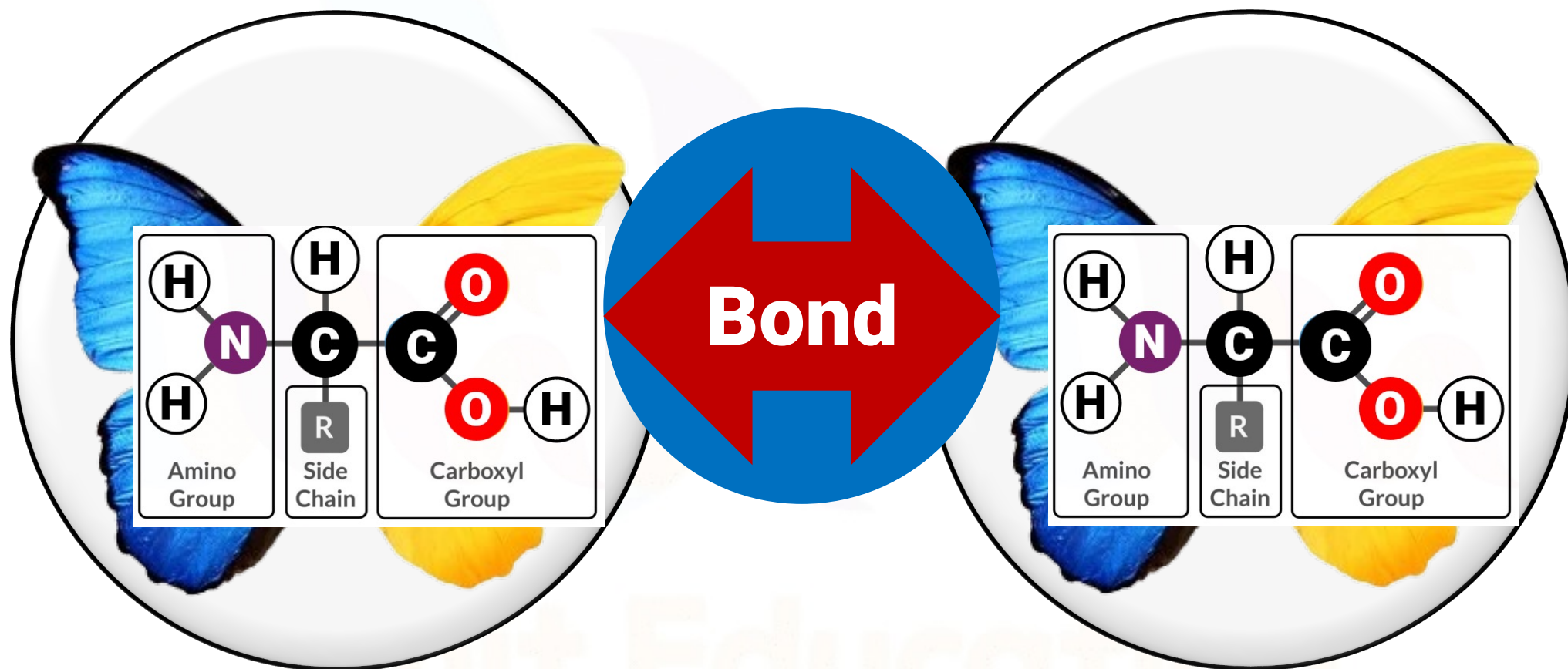
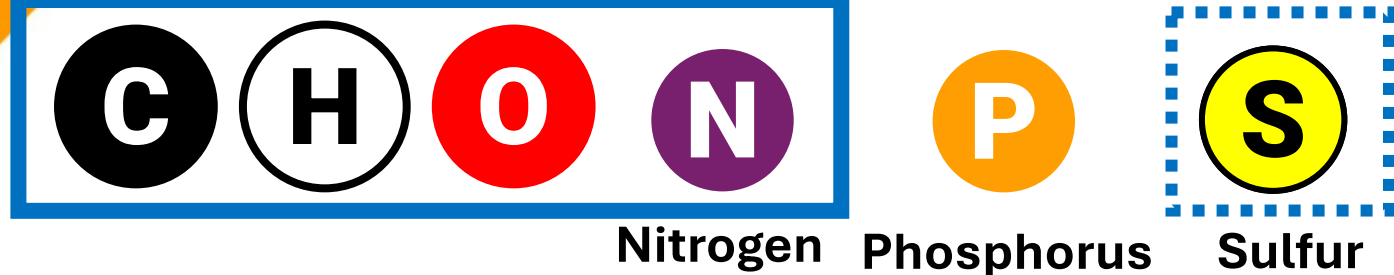
Phosphorus

Sulfur



Amino acids (胺基酸)

3. Proteins 蛋白質



A

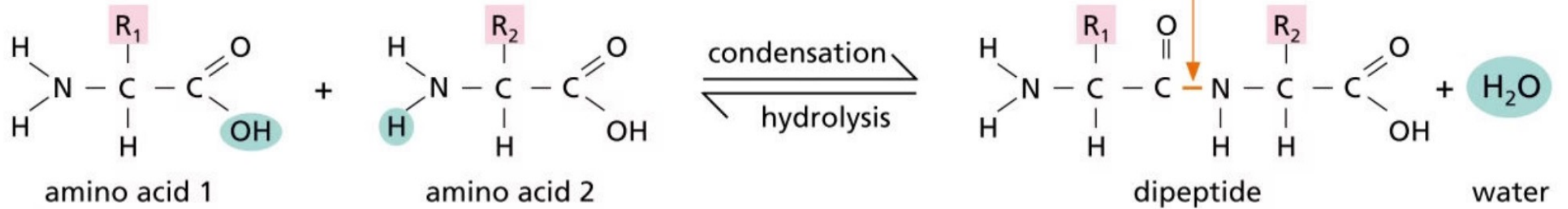
Dipeptide (二肽)

3. Proteins 蛋白質

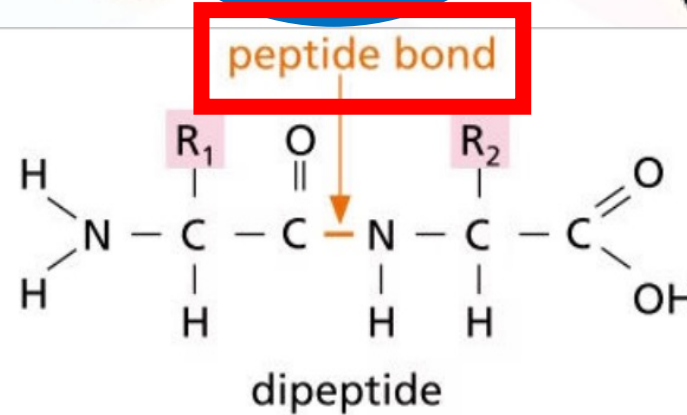
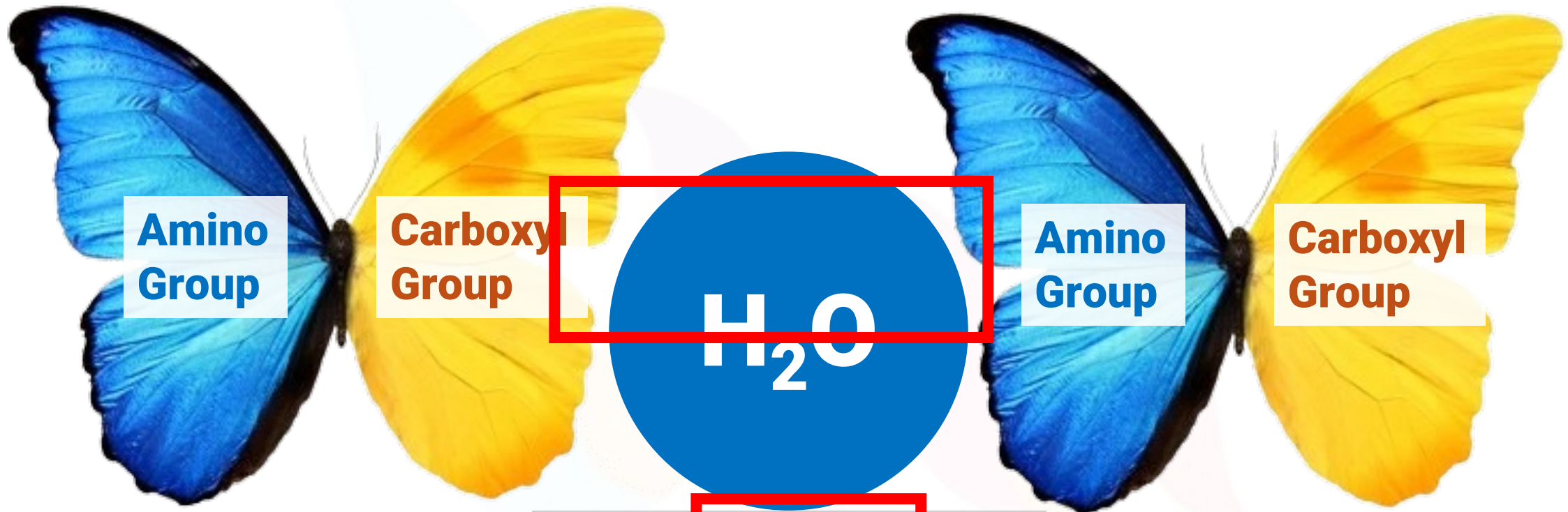
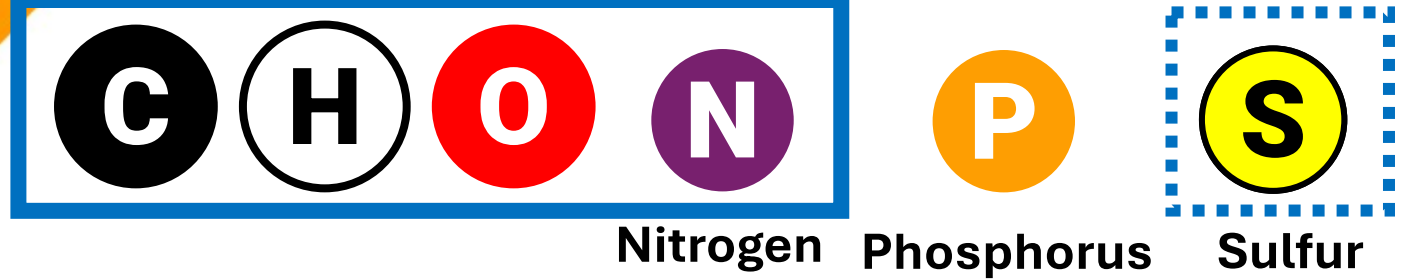


Phosphorus

Sulfur



3. Proteins 蛋白質

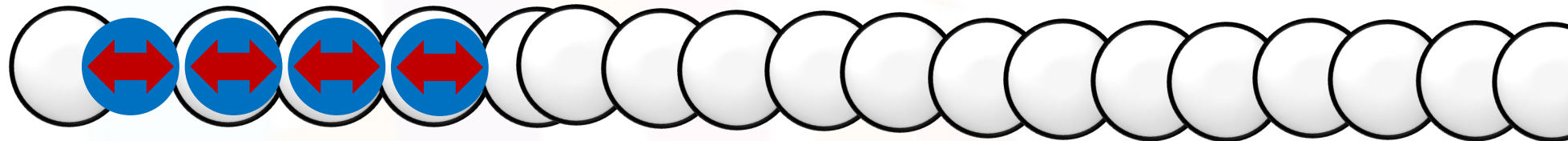
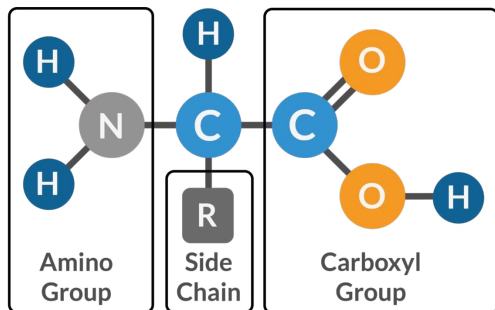


3. Proteins 蛋白質

Condensation 縮合

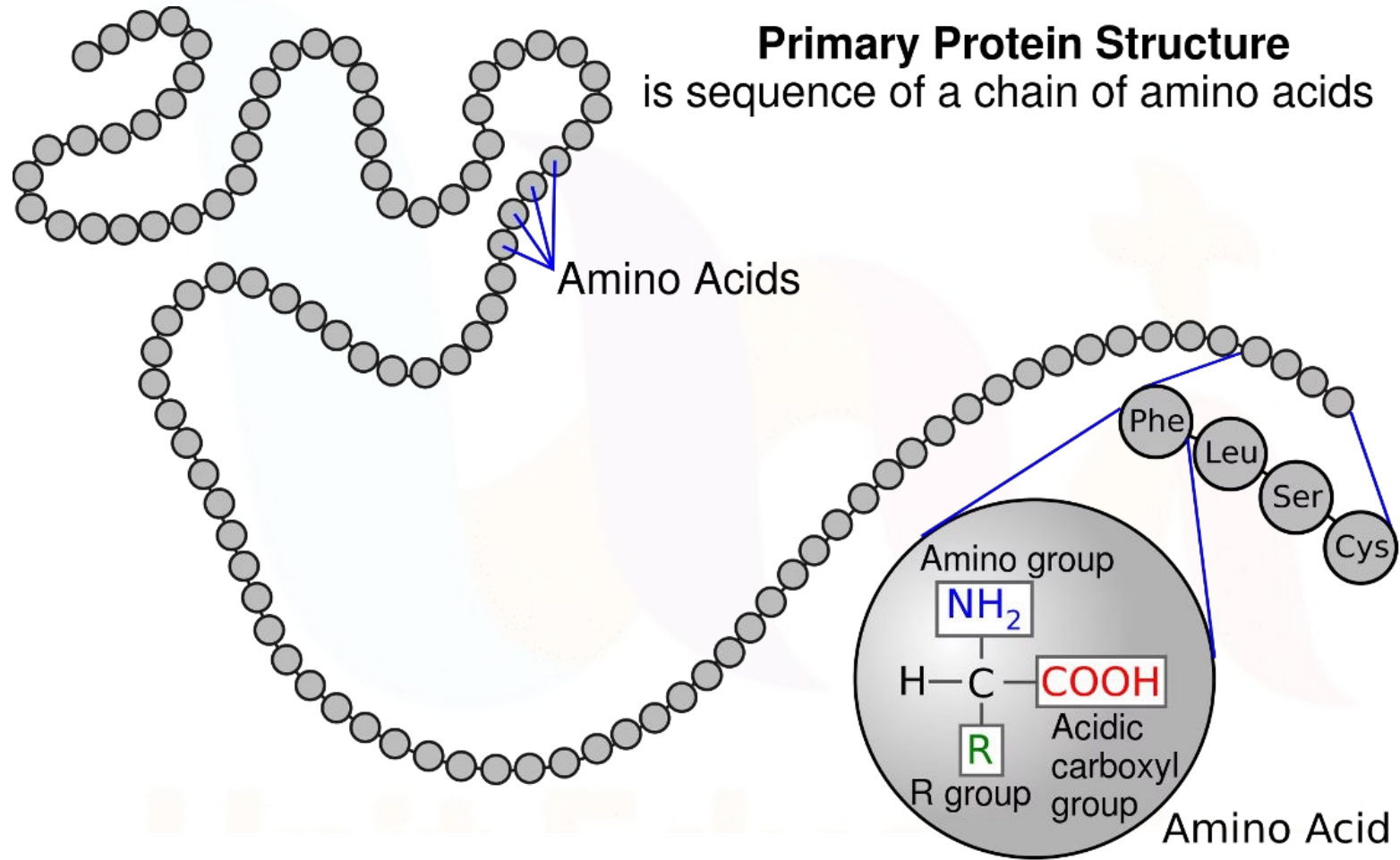


Amino Group **Carboxyl Group**



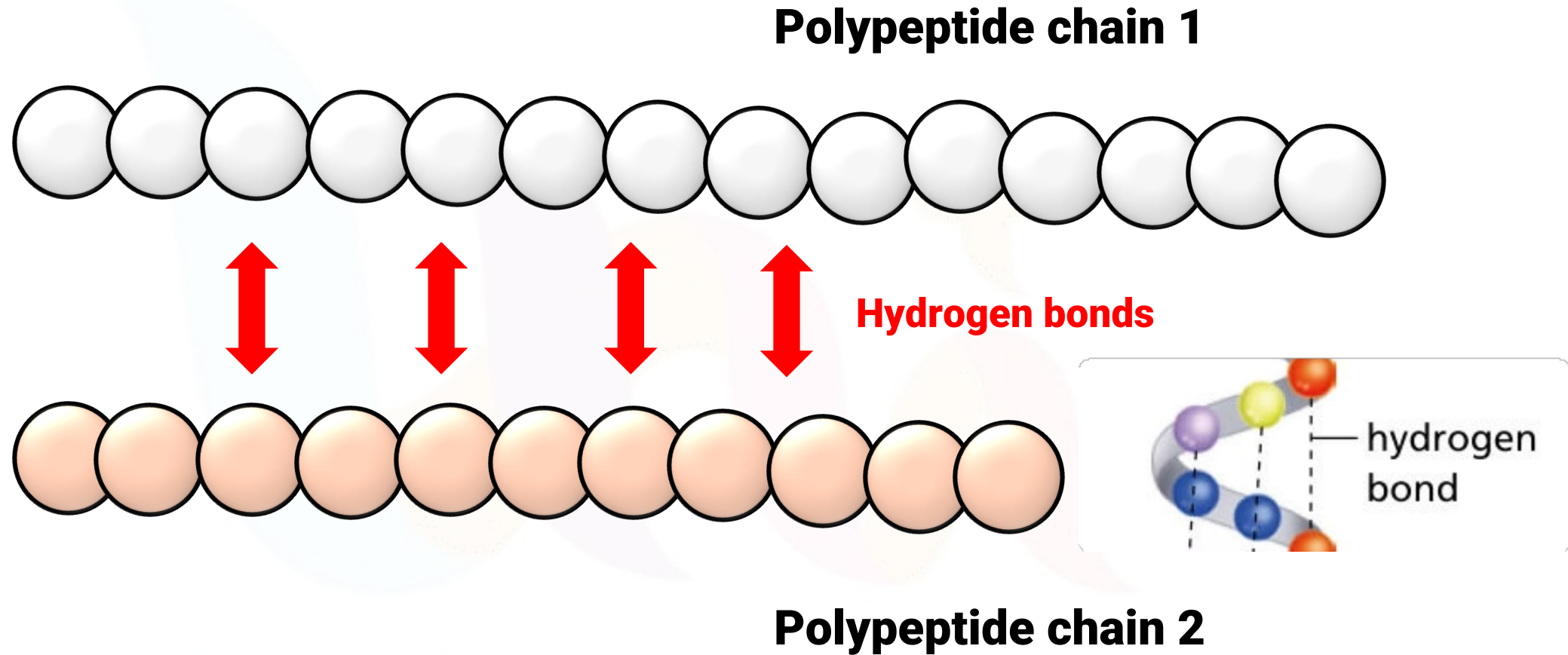
Polypeptide (多肽) Chain

3. Proteins 蛋白質



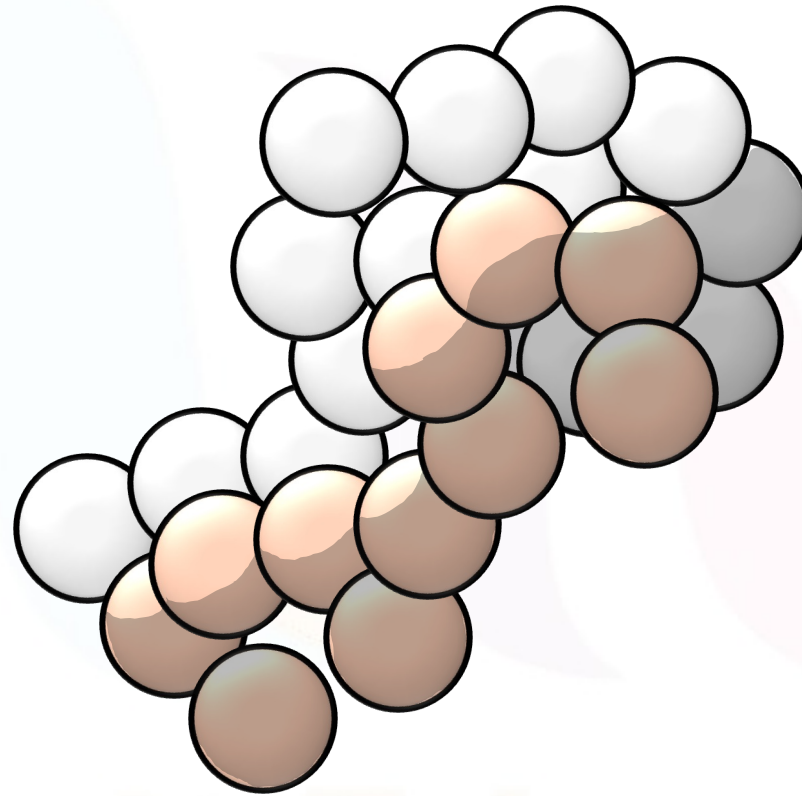
Polypeptide (多肽) Chain

3. Proteins 蛋白質



Polypeptide (多肽) Chains

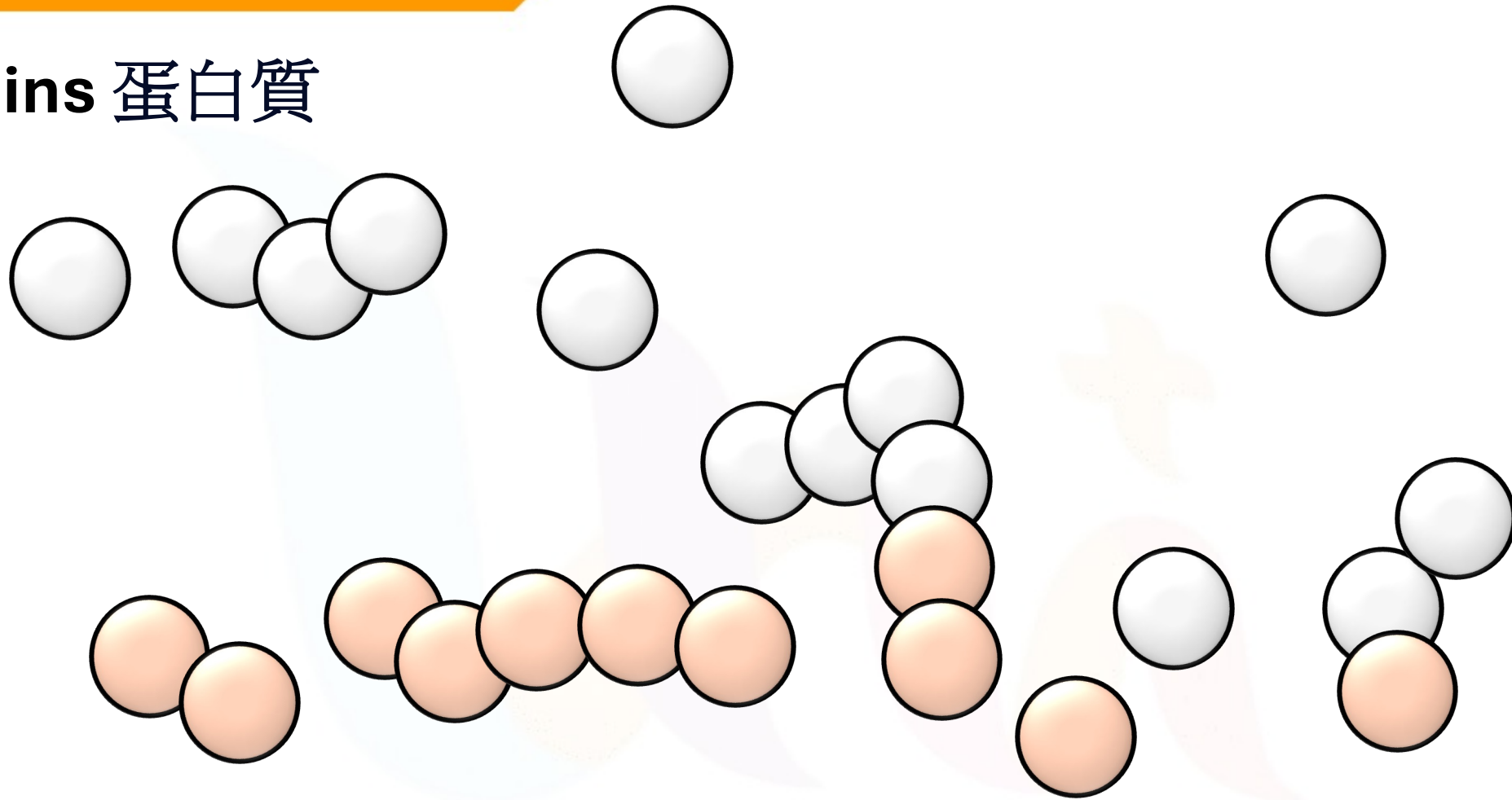
3. Proteins 蛋白質



**3D conformation
determine protein
functions**
For the rise
to 3D conformation

Proteins 蛋白質

3. Proteins 蛋白質



Suboptimal pH and temperature denature proteins